

Realtime Quranic ASR via a Kaldi-Style Hybrid

Architectural advantages, implementation roadmap, and time-budget analysis for replacing transformer ASR with a CTC + WFST pipeline in the Quranic recitation domain.

ABOUT THIS PROJECT

This is a personal endeavor to help create a new class of applications that advance new Muslims' understanding of and connection to the deen. The project is completely free and open source. Collaborators are warmly welcomed.

Abstract

General-purpose transformer ASR systems such as Whisper carry an architectural tax — fixed 30-second encoder padding, autoregressive token decoding, open-vocabulary output — that is wasted, and arguably harmful, when the input domain is Quranic recitation. The Quran is a closed textual corpus of 6,236 ayat with highly regular prosody (governed by tajwid) and a narrow style space dominated by the Hafs 'an Asim recitation. We argue that a pre-transformer “Kaldi-style” hybrid — a small CNN/Conformer acoustic model trained with CTC loss, decoded against a weighted finite-state transducer (WFST) compiled offline from the canonical Uthmani text — is the correct architecture for this problem. Such a system can achieve sub-100 ms end-to-end latency, ~35MB on-disk footprint, and near-perfect accuracy on Quranic speech, while being structurally incapable of hallucinating non-canonical text. We outline a phased implementation reaching a working prototype in 8–11 weeks of focused engineering.

This v2 incorporates lessons learned from a prior attempt at training ONNX-format Quranic ASR models on studio recitations augmented with synthetic masjid acoustic effects. That approach failed to generalise to actual imam recitation in the field — a textbook *sim2real* failure driven primarily by reciter-style mismatch (studio qari prosody is acoustically different from local imam recitation), compounded by augmentation that captured only a narrow slice of real masjid acoustics. The v2 design treats this as a binding methodological constraint: **fine-tune from a self-supervised multilingual pre-trained backbone, evaluate exclusively against real imam recordings from day one, and use real impulse responses rather than synthetic reverb if any augmentation is applied at all.**

Executive summary

- Whisper-base, our current production engine, has a structural latency floor of **~1.2–2.5sec/chunk** on phone CPU, regardless of input duration. This floor cannot be removed without retraining the model.
- For Quranic recitation specifically, this generality is not just unnecessary; it is *actively damaging*: Whisper hallucinates non-canonical Arabic on religious content, producing output that is worse than no

transcription.

- A constrained-domain alternative — CTC acoustic model + WFST decoder over Quranic-only paths — is structurally incapable of producing non-Quranic text and runs an order of magnitude faster.
- Estimated effort to working prototype on Android: **~7–10 weeks** for a single experienced developer — revised honestly from v0.3's 5–7 weeks after rejecting an in-app audio-collection pillar that contradicted the project's offline / privacy-first commitment. The data-prep collapse from Hetchy's *everyayah-phonemes* still saves ~2 weeks vs. v0.2, but Phase 0 (real-imam corpus collection by the project team, never from app users) adds ~2–3 weeks back. Net improvement vs. v0.2: ~1–3 weeks.
- **Architectural commitment (load-bearing, not decorative):** the application is **fully offline** — no internet required for any inference path — and **privacy-first**. User audio is never written to durable storage, never uploaded, never used to train any model the project ships. Every training corpus is collected by the project team or contributed via explicit, separately-consented channels. This constraint shapes the data-acquisition strategy in §5 and §7.
- **Competitive landscape (May 2026):** two non-empty positions exist. *Tarteel AI* leads research at ~4% Quranic WER but holds best models closed; their open *whisper-base-ar-quran* (5.75% WER, Apache 2.0) is the primary benchmark to beat. *RecitID* ships the consumer “Shazam for Quran” experience (iOS + Android, \$9.99/mo, ~18s identification latency, **cloud-only — recognition requires internet**). Neither incumbent serves the **fully-offline, free, open-source, mobile-first** lane this project occupies, and Tarteel cannot enter it without breaking their own business model.
- **The binding technical risk is the studio→masjid acoustic gap** (§5). Every public Quranic audio dataset — EveryAyah, *everyayah-phonemes*, Tadabur, Tarteel's closed sets — consists of professional qari studio recordings. None contain real masjid imam audio. A previous attempt to bridge this gap with synthetic-reverb augmentation failed to generalise. v0.4 treats real-imam data as a pre-Phase-1 deliverable, not a §5.2 footnote.
- **Sim2real lesson from prior work:** a previous attempt to train Quranic ASR models on studio recitations + synthetic masjid effects did not generalise to real imam audio. v2 mandates fine-tuning from a multilingual self-supervised backbone (Wav2Vec2-XLS-R, MMS-1B, or NeMo Conformer-CTC) and evaluating only against real masjid recordings from the outset.
- **Runtime: ONNX.** The acoustic model ships as ONNX, executed via ONNX Runtime Mobile. Pre-trained Arabic CTC backbones are widely available in this format, dramatically reducing the from-scratch training burden.
- This is the technical realisation of the “QuranicASR post-processor” IP angle previously identified: a defensible, mobile-first, religion-domain ASR system that no major lab is meaningfully pursuing.

1. The architectural tax of transformer ASR

1.1 Why Whisper has a fixed latency floor

Whisper's encoder always operates on a 30-second log-mel spectrogram (3000 frames at 100Hz, downsampled to 1500 audio “tokens”). This shape is hardcoded into the architecture by way of **learned absolute positional embeddings** — 1500 trained vectors that the encoder weights co-depend on. Inputs shorter than 30s are zero-padded; the encoder runs the full 1500-position self-attention regardless of how much real audio it received.

The decoder is autoregressive and emits one token at a time, each step computing self-attention over its own growing output and cross-attention over all 1500 encoder embeddings. Translation is not a separate decoding pathway — it is the same decoder writing English vocabulary tokens. The end-to-end inference cost on a modern phone is dominated by the encoder (constant) plus the decoder (linear in tokens emitted). Empirically, this floors at roughly:

Whisper variant	Disk	Phone-CPU latency / 6s clip	Arabic WER
tiny	39 MB	~0.4 s	40–60% (unusable)
base (current)	142 MB	~1.2–2.5 s	20–30%
small	488 MB	~3–5 s	12–18%
base, q5 quant	57 MB	~0.8–1.5 s	≈ base

Table 1. Whisper latency/accuracy on Arabic, base architecture.

1.2 Failure modes specific to Quranic content

- **Hallucination of non-canonical text.** Whisper's decoder is trained to produce coherent Arabic, not *specifically Quranic* Arabic. When the audio is recited Quran, it tends to emit a phonetically-close but textually-wrong approximation, often with broken tashkeel.
- **Tashkeel collapse.** Quranic text is fully-vowelled (with diacritics). Whisper's Arabic vocabulary is partially un-vowelled, so even when the consonantal skeleton (rasm) is correct, the diacritics drop or scramble.
- **Unrecoverable mistranslations.** Translation pass is generative; an incorrectly-decoded ayah produces a confidently-stated wrong English meaning, which is worse than emitting nothing.
- **Religious-sensitivity asymmetry.** A 10% word-error-rate is acceptable on a podcast and unacceptable on a verse from scripture. The cost of error is categorically higher than for general speech.

2. The constraint structure of Quranic recitation

Speech recognition is a search problem: among all possible character sequences, find the one that best explains the audio. The transformer's strength is handling *open-vocabulary* search. Quranic recitation collapses every dimension of that search to a near-finite domain:

Dimension	General Arabic ASR	Quranic recitation
Vocabulary	Open (~10 ⁶ word forms)	6,236 ayat, fixed
Output token space	Whisper: 51,865 BPE	~50 phoneme/letter classes
Prosody	Free (any speaker, any style)	Tajwid-governed (regular)
Speaker style	Open	Mostly Hafs ‘an ‘Asim
Language model	Required (huge)	The Quran itself — no probabilistic LM needed
Translation target	Generative	Selected canonical translation (Sahih, Pickthall, etc.)

Table 2. Search-space collapse from general to Quranic ASR.

The output token-space reduction alone is roughly three orders of magnitude. More importantly, the **language model collapse** — the fact that the set of valid output sequences *is exactly the Quran* — means we no longer need a neural language model at all. We need a graph.

3. Architectural foundations

3.1 Connectionist Temporal Classification (CTC)

CTC is a per-frame, parallel-emit alternative to autoregressive decoding. The acoustic model emits, for each ~20ms frame of audio, a probability distribution over a small label set augmented with a special *blank* symbol. After the entire audio has been processed, the output is collapsed: consecutive identical labels are merged, blanks are removed.

audio frames: [20ms][20ms][20ms][20ms][20ms][20ms]...

 v v v v v v

frame labels: 'b' '- ' 's' 's' 'm' '- ' -> 'bsm' (' - ' = blank)

Crucially, all frames are computed in a single forward pass. There is no autoregressive loop — each frame is independent at inference. Combined with a streaming-friendly encoder architecture (1D depthwise-separable CNN, or a Conformer with causal/chunked attention), this yields true real-time output: labels stream out as audio arrives, with sub-100ms latency.

3.2 Weighted Finite-State Transducers (WFST)

A WFST is a directed graph where nodes represent states and edges carry an input symbol, an output symbol, and a weight (cost). A path through the graph from start to end is a valid input/output sequence pair, with total cost equal to the sum of edge weights traversed. In classical Kaldi pipelines, the decoder graph is a composition of four transducers:

HCLG = H o C o L o G

```

|      |      |      +-- G : language model (n-gram or graph)
|      |      +-- L      : lexicon (word -> phone sequences)
|      +-- C      : context-dependent phone topology
+-- H      : acoustic-state topology (HMM or CTC)

```

For the Quran, the L■G stages collapse into a single, simple transducer: a flat enumeration of every legal phoneme path through the canonical text. There is no probabilistic n-gram — the Quran is exactly the set of allowed outputs. The graph is built once, offline, and shipped with the app.

3.3 Decoding: composing CTC with the FST

At inference, CTC posteriors are scored against the WFST via standard frame-synchronous beam search:

```

for each frame t:
  for each active hypothesis h in beam:
    for each outgoing edge e in FST from h.state:
      new_score = h.score + ctc_log_posterior[t][e.label]
                      + e.weight
      extend hypothesis h along edge e with new_score
  prune beam to top-K by score
after final frame: return best path

```

Because every hypothesis is constrained to follow a path that exists in the FST, the system **cannot** emit a sequence that isn't Quranic. This is not a soft preference; it is a structural invariant of the decoder. Hallucination of non-canonical text is mathematically impossible.

4. Proposed system architecture

The full pipeline, from microphone to displayed canonical text, is shown below. Each stage is sized to fit comfortably on a mid-range Android phone.

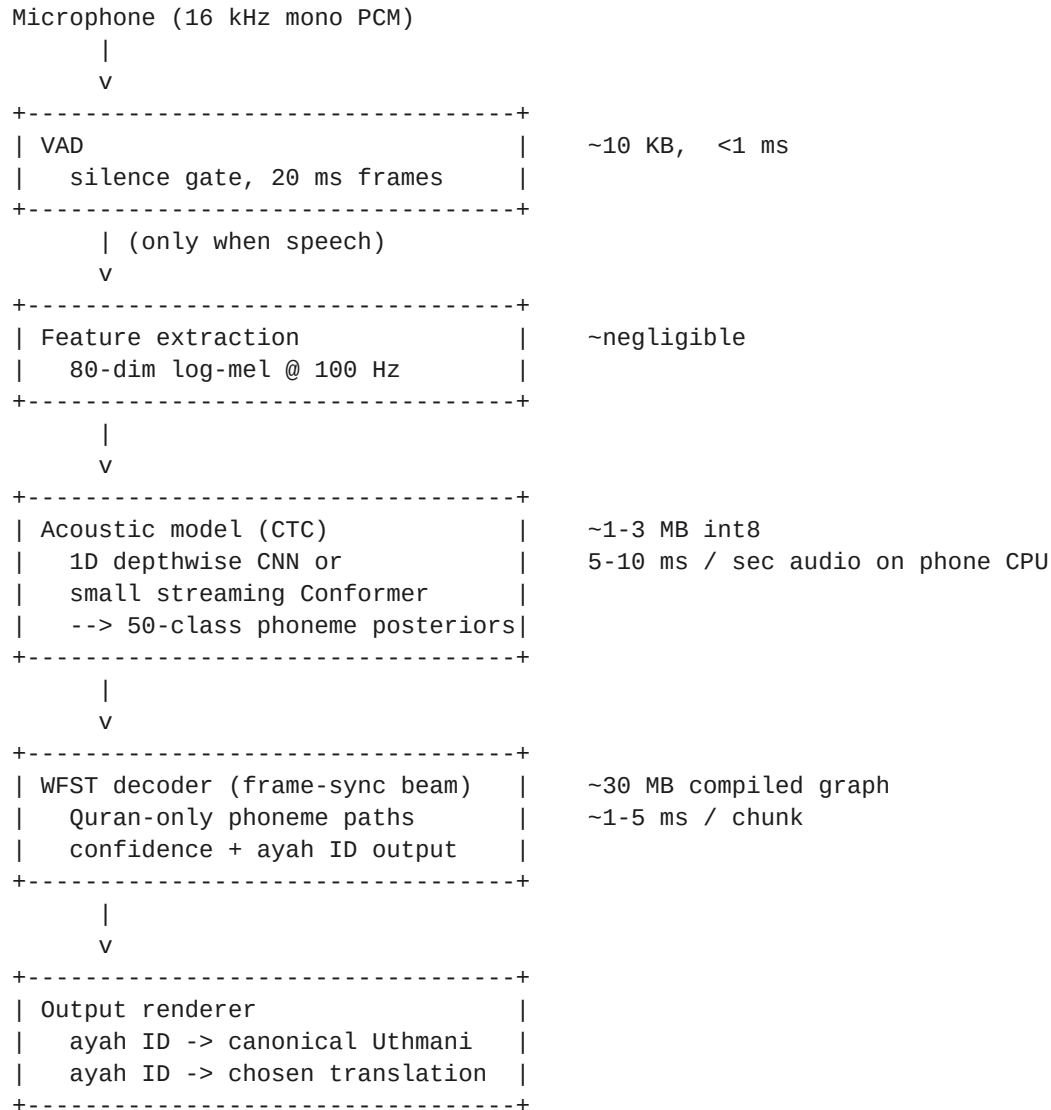


Figure 1. Realtime Quranic ASR pipeline.

4.1 Memory and latency budget

Component	Disk	Peak RAM	Latency
VAD	10 KB	~50 KB	<1 ms
Feature extraction	—	~2 MB	<1 ms
Acoustic model (int8 CNN)	1–3 MB	~10 MB	5–10 ms / sec audio
WFST decoder + graph	30 MB	~50 MB	1–5 ms / 1-sec chunk

Output / render assets	5 MB	—	—
Total	~35 MB	~60 MB peak	<100 ms end-to-end

Table 3. Resource budget on a mid-range Android device. For comparison, our current Whisper-base pipeline uses 142 MB on disk, ~250MB peak RAM, and 1.2–2.5sec per 6-second chunk.

5. Lessons from prior art

Earlier work on this project pursued a Quranic ASR model trained on studio recitations of professional qaris, with synthetic masjid acoustic effects (reverb, ambient noise) added as augmentation. The trained model performed well on held-out studio audio but **failed to generalise to actual imam recitation** recorded in real masjids. This is a textbook *sim2real* gap, and it carries hard constraints into the v2 design.

5.1 Diagnosis

Three factors, in decreasing order of likely contribution:

- **Reciter-style mismatch (dominant).** Studio qaris (Sudais, Husary, Mishary, Shuraim, etc.) have a specific prosody signature: slow tempo, maximum madd duration, crisp tajwid execution, projected trained voice. Local masjid imams recite faster, with shorter madds, looser tajwid, and natural conversational voice quality. The model learnt the qari distribution as a tight prior; the imam distribution lives outside it. CTC's frame-level timing sensitivity amplifies this — tempo shifts misalign expected phoneme durations.
- **Augmentation captured only a narrow slice of real masjid acoustics.** Synthetic reverb approximates one degradation pattern; real masjid audio additionally contains PA-system nonlinear distortion, codec round-trip artefacts, hot-mic gain riding, non-stationary congregation noise (cough, response prayers, foot scuffs, HVAC), variable mic distance, and impulse-response variability across masjid architectures (small carpeted prayer hall vs. tiled marble dome). The model learnt to undo one synthetic noise pattern and got fooled by everything else.
- **Distribution shift through batch-norm statistics.** If the architecture used batch normalisation, its running statistics were fit to the augmented-studio distribution. Real-imam audio has different per-batch statistics, and BN parameters do not adapt at inference.

5.2 Methodological non-negotiables for v2

These are not preferences; they are constraints required to avoid repeating the prior failure mode.

- **Real-imam evaluation set, established before any training.** A 30–60 minute held-out test set of real masjid imam recitation, hand-aligned to canonical text, must exist before Phase 2 begins. Every architecture, augmentation, and hyperparameter decision is evaluated against this set, never against synthetic data alone.
- **Real-imam *training* corpus, in addition to evaluation.** 10–30 hours of diverse real-masjid imam audio, force-aligned with *tarteel-ai/whisper-base-ar-quran* as the alignment oracle. Diversity across imams and masjid acoustics matters more than total hours.
- **Pre-trained self-supervised backbone, never train from scratch.** Self-supervised models (Wav2Vec2, HuBERT, MMS, etc.) pre-trained on thousands of hours of diverse voice data carry a much broader acoustic prior than any from-scratch training set could. The task becomes Quranic *fine-tuning*, not Quranic *training*. This is the single most important architectural consequence of the v1 *sim2real* failure.

- **If augmentation is used, prefer real impulse responses over synthetic.** Record IRs from 5–10 actual masjids with diverse architectures (clap, balloon pop, sine sweep). Pair with the MUSAN noise dataset for ambient. Add codec round-tripping (Opus low-bitrate, AMR) to capture phone-recording artefacts. Synthetic reverb alone is documented to be insufficient.
- **Layer norm, not batch norm.** Eliminates the BN-statistics-mismatch failure mode entirely. Self-supervised backbones generally already use layer norm.
- **Confidence calibration on real-imam audio, not on studio audio.** The WFST decoder's posterior threshold for substituting canonical text vs. deferring to Whisper must be tuned against the real-imam evaluation set. Calibrating on studio audio would produce overconfident substitution at deployment, exactly the failure mode the §5 sim2real lesson warns against.

5.3 Why public datasets do not solve this

Section 7's bootstrap leverage notwithstanding, every public Quranic audio dataset — EveryAyah, *hetchyy/everyayah-phonemes*, Tadabur, *tarteel-ai/everyayah*, the QUL recitation library, and Tarteel's closed proprietary sets — consists of **professional qari studio recordings**. None contain real masjid imam audio at scale. Even Tarteel's reported ~4% WER is on Quranic Arabic as *recited by qaris*, not as recited by the local imam during Salat al-Jumu'ah. The deployment distribution this project targets is structurally **not in any public dataset**, which is why real-imam data acquisition cannot be deferred or outsourced.

5.4 Data acquisition strategy — and what it explicitly is not

The training corpus and evaluation corpus are collected exclusively by the project team and explicitly-consenting external contributors. The application itself never collects, transmits, or uses user audio for any model-training purpose, ever. This is an architectural commitment, not a current-version limitation. Any feature that would require it is rejected at design time.

Three permissible sources, all of which preserve the offline / privacy-first commitment:

- **Project-team field recording.** The project team and volunteers record real-masjid audio at consenting partner masjids. Consent is explicit, written, and per-imam. The resulting corpus is released as a **public open dataset** (the inverse of Tarteel's closed-best-models posture) under a permissive license — a contribution that strengthens the open ecosystem regardless of what this project itself ships.
- **Already-public khutbah audio.** Imam podcasts, masjid live-streams, and Friday prayer broadcasts (including from Mecca and Medina) collectively represent thousands of hours of consent-implicit-by-publication content. Per-channel license verification is required, but the upper bound on collectable audio is very large.
- **Partner data sharing.** Many masjids already self-record khutbahs for archive or live-stream. Existing Quranic-ASR researchers (Hetchy, the Tadabur team, others) and open-source Quran initiatives may share unpublished real-masjid data under research/educational license. The Khutbah Translator parent project is a natural collaborator-attractor in this category.

Explicitly rejected: any pattern involving the deployed application uploading audio, gradients, or model deltas off-device for training, telemetry, or any other purpose — even with explicit user opt-in. The architectural guarantee “your audio never leaves your phone” is a load-bearing promise of this project, not a

configurable preference. A future on-device adaptation feature (operating purely in RAM, with audio discarded immediately and adapter weights stored locally only) is permissible *specifically because* it never violates this guarantee — not as an exception to it.

6. Runtime and deployment: the ONNX strategy

The acoustic model needs a runtime: a software engine that loads the trained model and runs forward passes on phone CPU efficiently. This decision is orthogonal to the architecture (the WFST decoder remains a separate C++ component regardless), but it materially affects training-pipeline ergonomics, the catalogue of pre-trained models we can fine-tune from, and binary size.

6.1 Runtime comparison

Runtime	Strengths	Weaknesses
ONNX Runtime Mobile	Best framework interop (PyTorch/NeMo & ONNX is the line); broad operator coverage, making Conformer	High line; broad operator coverage, making Conformer
TFLite	Mature on Android; tight Google integration.	Operator gaps for newer architectures; PyTorch&ran
NCNN	Tiny binary; very fast on ARM.	Narrower operator support; smaller ecosystem.
ggml (Whisper.cpp)	Already integrated for the v1 Whisper pipeline.	Designed for transformer LLMs; weaker streaming s

Table 4. Mobile inference runtime comparison.

For the Quranic acoustic model specifically, **ONNX Runtime Mobile is the right call**. The reason cascades from the methodological non-negotiable in Section 5.2: we are fine-tuning from a pre-trained self-supervised backbone. Every viable backbone in this category — Wav2Vec2-XLS-R, MMS-1B, NeMo Conformer-CTC, distil-Wav2Vec2 — is published as a PyTorch checkpoint, trivially exportable to ONNX. The TFLite conversion path for these models is lossy or broken; the ggml port would require substantial custom work.

6.2 Pre-trained backbone candidates

Each of the following can be fine-tuned on phoneme-aligned Quranic recitation and quantised to int8 for ORT Mobile deployment. The selection trades accuracy ceiling against on-device size.

Backbone	Architecture	Params (pre-quant)	Notes
Wav2Vec2-XLS-R-300M	Self-supervised CTC	300 M	Many Arabic fine-tunes on HF; well-trodden ONNX p
MMS-1B	Multilingual self-supervised CTC	1 B	Covers MSA Arabic; broadest voice prior; quantises
NeMo Conformer-CTC small	Conformer + CTC, streaming-native	30 M	Smallest viable; designed for mobile; Apache 2.0.
Distil-Wav2Vec2 Arabic	Distilled CTC	~95 M	Pre-distilled for mobile; lower ceiling.

Table 5. Pre-trained CTC backbone candidates for fine-tuning.

Default recommendation for v2 prototype: **NeMo Conformer-CTC small** as the starting point. It is the smallest and fastest, streaming-native by construction, and Apache-licensed. If accuracy on the real-imam test set is insufficient, escalate to a quantised Wav2Vec2-XLS-R or MMS-1B.

6.3 Deployment pipeline

PyTorch / NeMo checkpoint (pre-trained backbone)
|

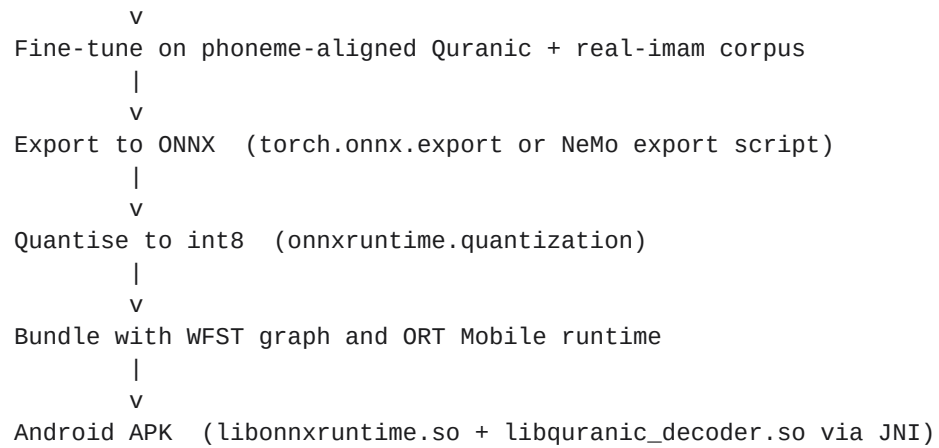


Figure 2. v2 model lifecycle from training to deployment.

7. Implementation roadmap

The work decomposes into **seven phases** — two real-imam corpus phases (0a, 0b) added in v0.4 to address the \$5 binding constraint, plus the five technical phases from v0.3. Total estimated effort is **~7–10 weeks** for a single experienced developer with part-time field-recording support. This is honest about the v1 sim2real failure: the real-imam corpus cannot be deferred. The everyayah-phonemes data-prep collapse still saves ~2 weeks vs. v0.2; Phase 0 adds back ~2–3 weeks; net improvement vs. v0.2 is ~1–3 weeks.

Phases 0a and 0b run in parallel with each other and partly in parallel with Phase 1. They are gating: Phase 2 (acoustic model fine-tuning) cannot begin until at least the evaluation corpus exists, and ideally also the training corpus. The *salat-quranlens* sister project's Whisper-Quran models remain useful as alignment oracles for non-EveryAyah audio.

Phase	Goal	Duration	Risk
0a. Real-imam evaluation corpus	Field record 30 h of real masjid imam recitation across 5–10 different mosques; 10 different m	1.5–2 weeks	Low
0b. Real-imam training corpus	Assign 10 h of real-masjid imam recitation. Source 2–3 h of field recording at partner masjids,	2–3 weeks	Low
1. Studio data prep	Use existing <i>hetchyy/everyayah-phonemes</i> dataset (1,248 h, 45 reciters, Tajweed-aware ph	1–2 weeks	Low
2. Acoustic model	Fine-tune from pre-trained backbone (NeMo Conformer-CTC, small as default; escalate to Wav2Vec2-XLS	1–2 weeks	Low
3. WFST construction	Tokenise full Uthmani Quran into phoneme strings (account for Tajwid rules: idgham, ikhfa, qalqala, etc). Compile v	1 week	Low
4. Mobile runtime	Bundle acoustic model (ONNX, int8) with ONNX Runtime Mobile (v32, ~500 MB). Implement compact fra	2–3 weeks	Low
5. Integration	Run as a parallel inference path alongside Whisper. Confidence scoring; substitute canonical text when QuranicM	1–2 weeks	Low

Table 6. Phased implementation plan.

7.1 Bootstrap leverage from existing assets

Phase 1’s force-alignment work is the traditional bottleneck for any CTC training pipeline (high-quality frame-aligned phoneme labels are scarce). For Quranic Arabic specifically, that bottleneck has just been eliminated by public work that did not exist when the v0.2 plan was written:

- **Hetchy / everyayah-phonemes** — 1,248h of EveryAyah + QUL recitation across 45 professional reciters, with automatically-generated Tajweed-aware phoneme labels under CC-BY-4.0. Provides train/dev splits and 16kHz mono audio. *This is exactly what Phase 1 was supposed to produce.*
- **Hetchy / Quranic-Phonemizer** — the open-source phonemizer that produced the above labels. Lets us extend the inventory to additional audio sources (for example, real-imam recordings) using the same phoneme vocabulary as the training set.
- **Faisal / tadabur** — 1,400h, 600+ reciters, WhisperX-aligned (April 2026 release). Useful for ablation comparisons and scale studies, but the CC-BY-NC-4.0 license means it cannot be used to train weights destined for commercial downstream applications.
- **tarteel-ai / whisper-base-ar-quran** (Apache 2.0, 5.75% WER, 12,970 monthly downloads) — a high-quality alignment oracle for any non-EveryAyah audio that needs labelling, in addition to functioning as the primary benchmark to beat.

- **salat-quranlens Whisper-Quran models** — ggml-quran-base.bin, ggml-quran-tiny-mosque-v2.bin, ggml-quran-tiny-masjid-v3.bin — remain useful as additional alignment oracles, particularly for any masjid-acoustics-augmented audio.

Net effect: Phase 1 collapses from a 2–3 week curate-and-align project to a 2–5 day evaluate-and-augment task.

7.2 Required infrastructure

- Single GPU (one A100 rented for ~20–40 GPU-hours suffices for phase 2 *fine-tuning* — substantially less than from-scratch training).
- OpenFst toolkit (open-source) for graph compilation.
- PyTorch + HuggingFace Transformers, or NVIDIA NeMo, for backbone fine-tuning.
- ONNX Runtime + onnxruntime.quantization for mobile export and int8 conversion.
- Kaldi's decoder libraries (Apache 2.0 license).
- Existing project Android infrastructure (already in place).
- **Real-imam evaluation corpus** (30–60 min hand-aligned, Phase 0a).
- **Real-imam training corpus** (10–30h, force-aligned, Phase 0b).
- **Field-recording kit for IR capture**: small balloon/clap source + portable recorder. Used at 5–10 partner masjids during Phase 0a/0b fieldwork to double as impulse-response collection (§5.2). One-time ~\$300 hardware spend.
- **Partner masjid coordination capacity** — written consent process, scheduling, ~1 day per masjid for IR + recitation recording.
- **License-verification process** for already-public khutbah audio sourcing (§5.4). Spreadsheet + per-channel/per-broadcaster review.

8. Real-time latency analysis

“Real time” in this context means end-to-end latency from a phoneme leaving the speaker's mouth to it being rendered on the listener's screen. The target is **under 200ms**, and ideally under 100ms, which is well below the threshold of perception for “live”. Whisper's lower bound is roughly **5–7 seconds** for our 6-second chunk window. The CTC+WFST design's lower bound is set by two factors: the CTC chunk size (typically 20–80ms of look-ahead) and the acoustic-model forward pass.

Stage	Whisper-base (current)	CTC + WFST (proposed)
Chunk fill time	5–6 s (full window)	20–80 ms
Acoustic forward pass	~1 s (encoder, fixed)	5–10 ms / sec audio
Decoding pass	0.5–1.5 s (autoregressive)	1–5 ms (frame-sync)
Network / disk I/O	0	0 (offline)
End-to-end p50	~5–7 s	<100 ms
End-to-end p99	~10 s	~150 ms

Table 7. Latency comparison, mid-range Android phone.

The proposed system is fast enough for use cases beyond the current post-the-fact transcript model: live closed-captioning during the khutbah itself, projected onto a screen for non-Arabic-speaking attendees, becomes viable. Whisper-base cannot support that use case at any model size.

9. Risk register

Risk	Likelihood	Mitigation
Sim2real gap repeats — fine-tuned model still fails Real-imam training corpus (Phase 0b), not just evaluation; real	Med	Diversify training set across reciters; include a small speaker-adaptation
Real-imam corpus collection underdelivers — part of Phase 0b, not just evaluation; real	Med	Diversify training set across reciters; include a small speaker-adaptation
Tajwid variation across reciters limits generalisation	Med	Diversify training set across reciters; include a small speaker-adaptation
Mixed audio (Quran interleaved with sermon speech) confuses the model	High	Run a parallel path to Whisper; substitute only on high posterior confidence
Less common qira'at (Warsh, Qalun) not handled	Med	v1 ships Hafs-only; add additional qira'at branches to FST in v2.
Female / child reciters under-represented in training data.	Med	Pitch-shift augmentation; targeted data collection from Tarteel-style recordings
WFST decoder proves slower than predicted on phone CPU.	Low	Beam-pruning aggressively; use Kaldi's decoder-faster; profile early
Phoneme inventory does not capture tashkeel sufficiently.	High	Adopt established Quranic phoneme set (~40 classes including madd
Integration UX confuses users (sudden text style change).	Low	Subtle 'verified' visual treatment; A/B test typography treatments

Table 8. Identified risks and mitigations.

10. Competitive landscape and project positioning

A May 2026 market scan (full inventory in the companion *MARKET_LANDSCAPE.md* document in this repository) reveals two non-empty positions in Quranic ASR, neither of which addresses the offline / free / open lane this project occupies.

10.1 Tarteel AI — the research leader

Tarteel reports state-of-the-art ~4% WER on Quranic Arabic. Their production stack uses NVIDIA NeMo + Riva, with recent optimisation work moving to **Wav2Vec2Bert + CTranslate2 + int8 + LitServe** — essentially the same architectural direction this whitepaper recommends, which is welcome external validation. They publish three open assets:

- **tarteel-ai/whisper-base-ar-quran** (Apache 2.0, 5.75% WER, ~12,970 monthly downloads, 161 likes on HuggingFace) — the most-used open Quranic ASR model. **The primary benchmark any new model must beat on real-imam audio.**
- **tarteel-ai/everyayah** dataset (26 reciters, word-level transcriptions, diacritised) — foundational corpus that powers most downstream Quranic ASR work, including the Hetchy phoneme-labelled derivative we use in Phase 1.
- **Quranic Universal Library (QUL)** — broader Quran resource library (text, translations, recitations, etc.).

Their best production models and proprietary audio datasets are deliberately *not* open-sourced — they are explicitly held back as “essential base resources for Tarteel’s sustainability.” This is reasonable and matters for our positioning: **Tarteel cannot enter the free / offline / open lane without breaking their own business model**, and they have no incentive to.

10.2 RecitID — the consumer product

RecitID (recitid.ai) is a polished iOS/Android consumer product that ships exactly the v2 vision — with one architecturally critical difference.

Feature	RecitID	This project (target)
Quran identification	Yes (surah / ayah / reciter from 200+)	Yes
Live khutbah translation	53 languages	Initially English, extensible
Inference location	Cloud-only	On-device
Internet required for recognition	Yes (“Audio detection and Live Mode require internet”)	No (fully offline)
Identification latency	~18 seconds	Target <100 ms
Pricing	Freemium — \$9.99/mo or \$59.99/yr for premium	Free forever
License	Closed source	Open source
Tech stack disclosed?	No (almost certainly cloud Whisper-class)	Public (CTC + WFST, ONNX runtime)

Table 9. RecitID vs. this project — comparable feature set, structurally different architecture and economics.

RecitID's existence and pricing tell us two important things: **(1) the demand is real** — consumers will pay \$60/year for this kind of tool; and **(2) the offline / free / open lane is structurally defensible** — RecitID cannot enter it without dismantling their subscription business. The lane and the mission framing (free public access for new Muslims) reinforce each other.

10.3 Where this project fits

	Cloud + Paid	Cloud + Open	Mobile + Free + Open
Identification only	RecitID (free tier)	—	(open lane)
Identification + Translation	RecitID Premium (\$60/yr)	—	← this project
ASR for memorisation / feedback	Tarteel app	Tarteel open Whisper baseline	(open lane)

Table 10. Market segmentation. The “Mobile + Free + Open” column is structurally unserved by either incumbent.

10.4 What makes this project defensible

- **Architectural choice of fully-offline operation** — ties hands of competitors locked into cloud business models. RecitID would need to dramatically rebuild to ship offline; Tarteel would lose subscription leverage.
- **Mission alignment with architecture** — the “free for new Muslims, no subscription, no internet required” framing is structurally consistent with the technical choices. Not a marketing veneer.
- **Mobile-optimised WFST runtime + ONNX integration** — a slim, drop-in C++ decoder library could be packaged as a standalone SDK for any Arabic-ASR-using app, not just ours. RecitID and Tarteel have no incentive to publish such a thing.
- **Mixed-audio confidence routing** — the integration logic that decides when to trust the Quran-specialised path vs. fall through to a general-speech path. Tunable, evaluable, and not currently shipped by any incumbent.
- **Open data / open weights commitment** — aligns with the natural collaborator pool (Hetchy's Quranic-Phonemizer is CC-BY-4.0; this project should match). Tarteel's closed-best-models posture leaves no path for external researcher contribution at the production frontier.

11. Conclusion

Whisper is not the right tool for Quranic recitation. Its architectural decisions — fixed-length encoding, autoregressive decoding, open-vocabulary output — are mismatched to a domain where the vocabulary is closed, the prosody is regular, and the cost of hallucination is high. A pre-transformer hybrid — the same family of designs that powered Google Voice and Apple Siri before 2019 — is the architecturally honest answer for this specific problem.

The proposed CTC+WFST system is faster (sub-100ms vs. ~5–7s end-to-end), smaller (35MB vs. 142MB), and more accurate on its target domain. It runs as a parallel path alongside our existing Whisper engine, substituting canonical text when it fires confidently and otherwise deferring to Whisper for the imam's general sermon speech. Implementation is tractable: **~7–10 weeks of focused work** (net improvement of

~1–3 weeks vs. the v0.2 estimate of 8–11 weeks), with the real-imam corpus phases honestly budgeted up front rather than deferred.

The market scan that informs this v0.3 also clarifies *why* this is the right project to build. Tarteel AI leads at the research frontier but holds best models closed. RecitID owns the consumer brand but is structurally cloud-only at \$60/year. **The free, fully-offline, open-source, mobile-first lane is unoccupied and unreachable for either incumbent without breaking their own business model.** That is the lane this project occupies, and the mission framing — helping new Muslims connect to deen at no cost and no friction — is the same statement said in non-technical language.

v0.4 closes one self-inflicted contradiction from v0.3: an in-app audio collection pillar that would have undermined the offline / privacy-first commitment. The corrected strategy treats **your audio never leaves your phone** as a load-bearing architectural promise, and acquires real-imam training data through project-team fieldwork, already-public khutbah sources, and partner data sharing — never from the deployed application. The resulting real-imam corpus is itself released as a public open dataset, strengthening the open ecosystem regardless of what this project alone ships.

“If your problem is constrained, your model should be constrained. The transformer is a hammer for an open-vocabulary world; the Quran is not an open-vocabulary problem.”

Document version 0.4 · Realtime-Offline-Quran-ASR project · v0.4 promotes the studio→masjid acoustic gap to the binding technical constraint (§5), adds Phase 0a/0b real-imam corpus collection to the roadmap, revises the timeline to ~7–10 weeks, and makes the offline / privacy-first commitment load-bearing — correcting a v0.3 in-app data-collection idea that would have undermined the project's core positioning. See companion MARKET_LANDSCAPE.md for the full open-resource inventory.